

TCMC • HARMONY EARS

# Chord Progression

Pitch Trainer

User Manual

(和弦進行音準練習)

App Version: v1.0

Producer

Taiwan Choral Music Center (TCMC)

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## 1. Overview

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"Chord Progression Pitch Trainer" is a web-based tool from the Taiwan Choral Music Center (TCMC) that lets you pick a common chord progression (such as I-IV-V-I, "6415," or "16251"), a key, and the voice part you want to practice (Soprano/Alto/Tenor/Bass), then sing your way through the whole progression along with a metronome while a real-time microphone check tracks your pitch.

This app uses the same tuning approach as the companion "Chord Voicing Trainer": each chord's root sits at standard 12-tone equal temperament, while its third, fifth and seventh are tuned in Just Intonation (pure ratios). The goal is to build a feel for the "locked-in" sound of in-tune chords, not just hitting the right note names.

### Key Features

- 16 built-in chord progressions, covering common major/minor progressions, dominant-7th/turnaround (vi7-ii7-V7) variants, and a "Golden Progression" that includes chord inversions and passing 7th chords
- Every progression can be sung in all 12 keys (all 12 have been checked and adjusted)
- Once a voice part is selected, each part has its own / toggle so you can choose which parts you hear — mute your own part once you know it
- An on/off metronome, adjustable tempo (40–160 BPM), and a loop toggle to repeat the whole progression
- A  "Voicing Table" showing, for every chord in the progression, each voice part's solfège syllable, note name, and offset from equal temperament in cents — tap any cell to hear that note
- Real-time microphone pitch detection with a  $\pm 6$  cent tolerance band, shown live on the chart
- Chinese / English interface toggle



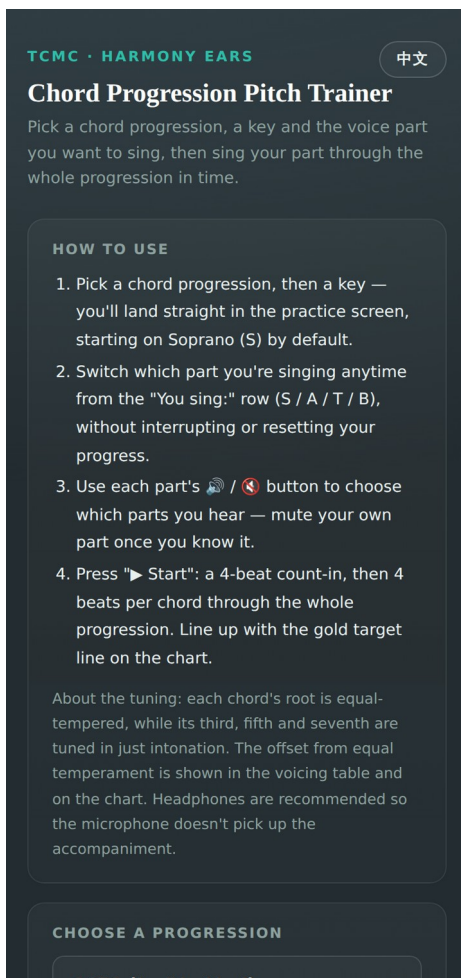
**Tip:** Headphones are recommended. Over speakers, the accompaniment can bleed into the mic and throw off the pitch judgement.

## 2. Interface Overview

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### Screen 1: Choose a Progression

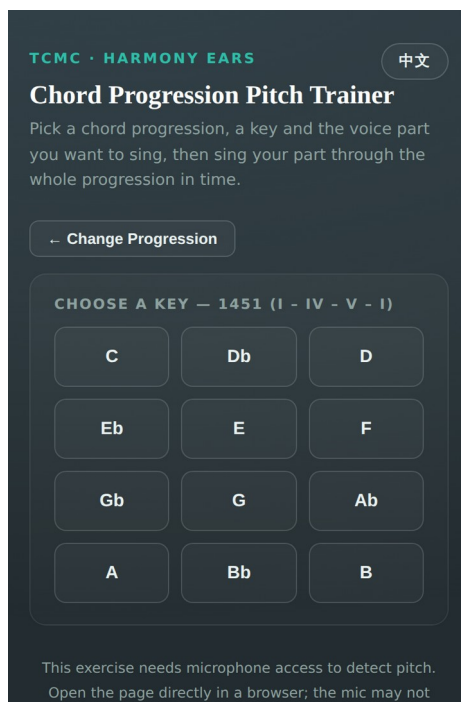
A short "How to Use" summary appears at the top, followed by a list of the 16 built-in chord progressions. Tap one to continue.



Screen 1: Choose a Progression

### Screen 2: Choose a Key

The 12 keys are shown as buttons — all 12 are available to choose from. Once you pick one you land straight in the practice screen — there's no separate voice-part step, since the practice screen's "You sing:" row lets you switch parts anytime.

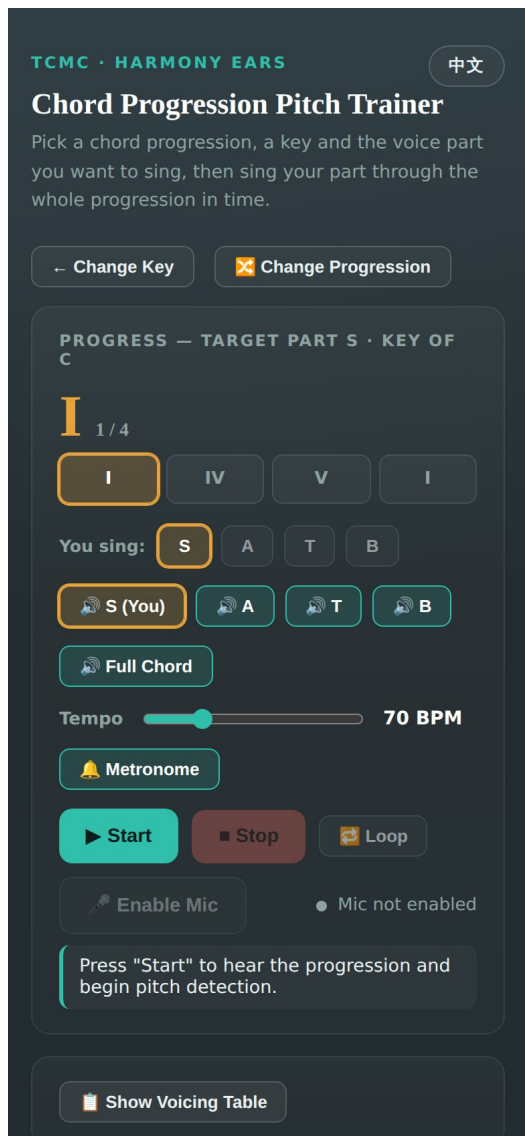


Screen 2: Choose a Key

### Screen 3: Practice Screen

This is the main practice screen — you land here right after choosing a key, starting on Soprano (S) by default. From top to bottom it contains:

- A progress row — every chord in the progression (roman numerals), with the current one highlighted in gold
- A "You sing:" row — tap to switch your target voice part (S / A / T / B) at any time, without interrupting or resetting your progress
- Per-part / toggles plus a "Full Chord" master toggle, controlling which parts you hear during playback
- A tempo slider and metronome toggle
- Start / Stop / Loop / mic toggle
- A Voicing Table (collapsed by default — tap to expand)
- The pitch chart with live readouts (current pitch, offset from target, judgement)



Screen 3: Practice Screen

### 3. How to Use

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1. From the first screen, pick the chord progression you want to practice.
2. Choose a key — you'll land straight in the practice screen, starting on Soprano (S) by default.
3. Tap the part you want to sing in the "You sing:" row (S / A / T / B). You can switch anytime — it never interrupts or resets your progress.
4. (Optional) Open the  Voicing Table first to preview every chord's solfège and note names for your part — tap any cell to hear it.
5. Use each part's  /  toggle to decide which parts you hear. While you're still learning it, leave everything on; once you know your part, mute it and rely on your ear and the pitch chart instead.
6. Adjust the tempo (BPM) and metronome toggle if needed.
7. Press " Start": a 4-beat count-in plays, then each chord plays for 4 beats (speed set by the tempo slider) through the whole progression. Your browser will prompt for microphone access — choose Allow.
8. Sing along, aligning with the gold target line on the pitch chart. Your trace shows green within  $\pm 6$  cents of the target, and red outside that band.
9. The session stops automatically once the progression finishes; with " Loop" enabled it restarts from the top automatically. You can also press "■ Stop" at any time.
10. To change your key or progression, use the "← Change Key" or " Change Progression" buttons in the top-left corner — no page reload needed.

 **Tip:** If you're using an in-chat or in-app preview window, browser security restrictions may block the microphone permission request — download this HTML file and open it directly in a browser instead.

## 4. Progressions at a Glance

There are currently 16 built-in progressions:

| Progression                                        | Description                                                                                  |
|----------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1451 (I - IV - V - I)                              | The most basic major-key cadence progression                                                 |
| 6415 (vi - IV - I - V)                             | A common pop-song progression                                                                |
| 16251 (I - vi - ii - V - I)                        | A common jazz/pop progression including the ii chord                                         |
| Cycle of Thirds (I-vi-IV-ii-vii°-V-iii-I)          | A descending cycle built on thirds                                                           |
| I - IV - I                                         | Basic progression                                                                            |
| I - V - I                                          | Basic progression                                                                            |
| I - V - IV - I                                     | Basic progression                                                                            |
| I - vi - I                                         | Basic progression                                                                            |
| I - iii - IV - I                                   | Basic progression                                                                            |
| i - iv - V - i (minor)                             | Natural-minor cadence with a major dominant                                                  |
| I - IV - I - V - I                                 | Basic progression                                                                            |
| i - VI - VII - i (natural minor)                   | A natural-minor progression                                                                  |
| 1451 V7 (I - IV - V7 - I)                          | The dominant-7th version of 1451                                                             |
| i - iv - V7 - i (minor)                            | A minor-key dominant-7th cadence                                                             |
| I - vi7 - ii7 - V7 - I                             | A jazz-style "turnaround" built with 7th chords                                              |
| Golden Progression (I-V6-vi-iii6(7)-IV-I6-ii(7)-V) | Features chord inversions and passing 7th chords, prioritizing smooth stepwise voice leading |



**Tip:** All 16 progressions have been verified to be singable in all 12 keys, with every voice part's notes falling within a comfortable range.

## 5. Voicing Table

Tap "📄 Show Voicing Table" to expand a table: columns are each chord in the progression (labeled with roman numerals), rows are the S / A / T / B voice parts, and your currently selected target part is marked with a 🌟.

Each cell shows:

- Large text: the solfège syllable (movable-do — Do is the tonic in a major key, La is the tonic in a minor key)
- Smaller text below: the note name (e.g. C4, Eb4)
- Top-right number: which chord tone this is (1 root, 3 third, 5 fifth, 6 sixth, 7 seventh)
- Below that: the offset from equal temperament in cents (e.g. +2¢, -14¢), so you can check the tuning direction before you sing

Tap any cell to hear that exact note immediately — handy for checking what you're about to sing before pressing Start. Tap a roman numeral in the top row to hear the whole chord at once.

📄 Hide Voicing Table

|     | I                             | IV                            | V                             | I                             |
|-----|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| S 🌟 | Sol <sup>5</sup><br>+2¢<br>G4 | La <sup>3</sup><br>-14¢<br>A4 | Sol <sup>1</sup><br>+0¢<br>G4 | Sol <sup>5</sup><br>+2¢<br>G4 |
| A   | Mi <sup>3</sup><br>-14¢<br>E4 | Fa <sup>1</sup><br>+0¢<br>F4  | Re <sup>5</sup><br>+2¢<br>D4  | Mi <sup>3</sup><br>-14¢<br>E4 |
| T   | Do <sup>1</sup><br>+0¢<br>C4  | Do <sup>5</sup><br>+2¢<br>C4  | Ti <sup>3</sup><br>-14¢<br>B3 | Do <sup>1</sup><br>+0¢<br>C4  |
| B   | Do <sup>1</sup><br>+0¢<br>C3  | Fa <sup>1</sup><br>+0¢<br>F2  | Sol <sup>1</sup><br>-0¢<br>G2 | Do <sup>1</sup><br>+0¢<br>C3  |

In each cell: solfège syllable (large) and note name. Top right: which chord tone it is (1 root, 3 third, 5 fifth, 6 sixth, 7 seventh) and, below it, the offset from equal temperament in cents. Tap any

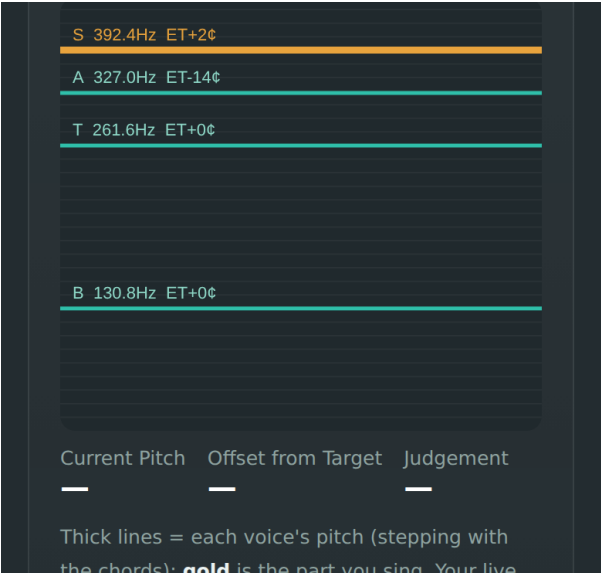
Voicing table example

## 6. Pitch Chart & Tuning Judgement

The chart's vertical range is fixed across the whole progression, so the reference lines visibly move up and down as chords change, rather than re-centering (and looking static) every time the chord changes.

The thick stepped lines mark each voice part's reference pitch; gold marks your current singing target. Once you start singing, your live pitch trace appears in green when within  $\pm 6$  cents of the target, and red when outside that band.

Below the chart you'll also see the detected frequency (Hz), the offset in cents from the target, and a text judgement (✓ In tune / Sharp / Flat).



Pitch chart example (before detection starts)

## 7. Troubleshooting

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### Microphone not responding / can't detect pitch

- Confirm your browser has allowed this page to use the microphone (check the permission icon in the address bar)
- If you're using an in-chat or in-app preview window, security restrictions may block the microphone — download the HTML file and open it directly in a browser instead
- Confirm the URL starts with HTTPS; a plain local file (file://) often can't get microphone access on some mobile browsers

### No sound

- Check that your device isn't muted and the browser tab hasn't been silenced by the system
- Some mobile browsers need one manual tap before they'll unlock audio playback — if there's no sound the first time, try tapping again

### Pitch trace jumps around or the judgement feels unstable

- Headphones are recommended over speakers, to avoid the accompaniment bleeding into the mic and throwing off detection
- Sing in a quiet environment and keep a consistent distance from the microphone

## 8. Version History

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Current version: v1.0

### Development Highlights

- v1.0: initial public release. All 16 chord progressions verified singable in all 12 keys; bilingual interface, voicing table, metronome and tempo control, and real-time microphone pitch detection.

### Production Team

Produced by: Taiwan Choral Music Center (TCMC)

Producer: Ray Chu (朱元雷)

Nature of software: a web-based tool for choral voice-part rehearsal and self-practice